

Contractor Safety Manual



JHABUA POWER LIMITED
Barela, Seoni (MP)

MISSION : Zero Harm
No Harm to any JPL Stakeholder

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1.0 INTRODUCTION/PURPOSE

- ❖ Jhabua Power Limited (JPL) is going to implement this Contractor Safety Manual to allow the contracting Organization to conduct the business at JPL as safely as possible. The manual is an agreement between JPL & the contracting agency to achieve the mission of Zero Harm.
- ❖ We **INSIST** that the contractor agency must familiarize his people with the contents of this manual and abide by all Rules and Guidelines mentioned there in the manual.
- ❖ Contractor shall be solely responsible for the safety of his employees, sub-contractors and agents during execution of the work

2.0 SCOPE

The JPL Contractor Safety Manual applies to all the Contractors providing services for Jhabua Power Limited.

3.0 DEFINITIONS

- JPL: Jhabua Power Limited
- The Factories Act: The Factories Act, 1948
- Contractor: Organization who is contracted by JPL to perform work or services for JPL at JPL premises.
- Subcontractors: Organization that is contracted by the Contractor to perform work or services for JPL at JPL premises.
- Contractor Employee: Any employee or service provider of a Contractor or Subcontractor.
- Side Incharge (Contractor Lead): The person designated by a Contractor and is responsible for Safety of the Contractor Employees.
- JPL Contact (EIC, Engineer in charge): The JPL employee who is responsible for JPL's contract with the Contractor. This responsibility may be designated, but the contact will be clearly identified prior to start of the work.

4.0 GENERAL RESPONSIBILITIES:

- JPL intends to hire only those Contractors who can provide services in a safe and healthy manner. By accepting a contract, the Contractor commits that he has the required skills, experience, expertise and commitment to perform work in a safe and healthy manner, and will allow only those Contractor Employees who are properly trained and supervised to work at JPL premises.
- If a contract between JPL and a Contractor allows for the Contractor to subcontract, the Subcontractor must have all the skills, experience, expertise and commitment to perform work in a safe and healthy manner, and that the Contractor will allow only those Subcontractor Employees who are properly trained and supervised to work at JPL premises. The Contractor is also responsible for providing this Contractor Safety Manual to the Subcontractor before work on JPL premises begins.
- A Contractor is responsible to keep JPL informed about its activities, and the activities of its Subcontractors. This is to be accomplished by the Side Incharge (Contractor Lead) with the JPL Contact.
- JPL is not responsible for safety and health policies or practices of any Contractor or Subcontractor. This Manual is to provide guidance to Contractors and Subcontractors on how they can satisfy their own

responsibilities in this regard.

5.0 SAFETY REQUIREMENTS:

- Contractor shall have sufficient knowledge, experience and understanding of thermal power plant work practices, safety & health hazards and other regulatory requirements pertaining to the work to be performed.
- Contractor shall perform the work using qualified workers who are adequately trained in the requirements of their particular job and skilled in the work assigned to them.
- Contractors shall provide proof of worker credentials (training, Qualification certificates, license etc.) on request/joining.
- Contractor shall comply with the requirement of Factory Act 1948, M P Factory Rules 1962 and other central & State laws, rules, regulation & time to time released orders of governing authority.
- Contractor shall comply with all company posted requirements, information provided by the company on site specific hazards and emergency response plans.
- Contractor shall review this manual with his employees, sub-contractors and consultants.
- Contractor shall have dedicated safety representative at the work site all the times. Contractor shall provide the qualifications of the proposed safety representative to company for review and approval.
- Contractor dedicated qualified safety representative shall perform inspection of operations, facilities and equipment's used in the performance of the work and participate in joint inspections, audits with company on request. Contractor shall immediately address any unsafe conditions, equipment's or action identified during inspection.
- Contractors shall ensure workers comprehend job specific safety related information including individuals in English, Hindi, Punjabi or any language which is easily understandable.
- Regularly Scheduled Safety Meetings: Contractor shall conduct regularly scheduled safety meetings. Attendance shall be required by all workers. Contractor shall keep a written record of the meetings that includes date, location, names or signatures of attendees, and topics covered. Contractor shall inform workers of factual circumstances resulting in incidents and near misses and discuss how to correct and prevent such situations from recurring.
- Daily Toolbox talk Meetings: Contractor shall conduct and document a daily morning safety meeting with all applicable workers to discuss Work activities, address any safety and health concerns for the Work to be performed, review any near miss incidents and how they could have been avoided, and prepare or review the appropriate Job Safety Analysis. Contractor shall provide such documentation to JPL upon request.

6.0 WORKPLACE REQUIREMENTS:

- Professional Conduct - Contractor shall conduct itself in a professional manner at all times while on Company Property. Horseplay, practical jokes, any type of harassment, abusive or objectionable language, or other inappropriate behavior on the job shall not be tolerated.
- Consequences for Non-Compliance - Working safely is a condition of employment at JPL. Any Contractor violating these rules and/or procedures will be required to permanently leave JPL premises.
- Weapons - All firearms, knives and other weapons are strictly forbidden at all JPL premises.

- Acts and threats of violence (physical or verbal) are strictly prohibited.
- Contractor shall inform its employees, suppliers, and subcontractors before entering Company Property that Company and its authorized representatives can search the person, vehicle, and other property of individuals while entering or departing from Company Property.
- The possession or use of narcotics, drugs, or intoxicating beverages of any kind is prohibited on JPL premises. Workman of Contractor shall immediately remove from the Workplace, if found in drug and alcohol testing violation.
- Contractor shall not bring unauthorized individuals (e.g., friends, relatives, or observers) onto Company premises.

7.0 SITE ACCESS

- Each Contractor Employee must carry his ID card when entering or providing services at JPL.

8.0 SAFETY PROGRAM

- The Contractor will have a safety program that outlines the requirements for performance of Contractor Employees specific to their activities. The Contractor is responsible for ensuring that this safety program meets the requirements of law, including but not limited to compliance with applicable Factories Act and other legal requirements.
- The Contractor will ensure that Contractor Employees have been trained prior to performing any activity at JPL
- The Contractor will ensure that its employees know the requirements outlined in this manual prior to beginning any work activity.
- The Contractor will communicate specific hazards found at JPL that may affect the safe work of Contractor Employees (e.g., working with chemicals, working in confined places, Electrical Hazards).
- The Contractor will be responsible for the direct supervision of Contractor Employees.

9.0 EMERGENCY RESPONSE AND EQUIPMENT

- Access to exits and to any emergency equipment (e.g., safety showers, eyewash fountains, firefighting equipment) must be kept clear at all times.
- The Contractor is responsible for communicating emergency procedures to Contract Workers. At all JPL facilities, immediate evacuation is required when audible alarm is sounded and/or an announcement is made to evacuate.
- Contractor Employees must leave by the closest/safest exit, as quickly as possible. After exiting the building, Contractor Employees must assemble in marked Safe Assembly Point and report immediately to the Side Incharge (Contractor Lead).
- The Side Incharge (Contractor Lead) is responsible for informing their JPL contact that the evacuation of all Contractor Employees was successful.

10.0 SPILLS AND INCIDENTAL RELEASES OF HAZARDOUS MATERIALS

- Contractor Employees must report any spill of chemicals or hazardous materials to their JPL contact immediately.
- A Contractor bringing any chemical or hazardous material onto JPL premises is required to provide their JPL contact with an MSDS for each substance and to use only approved, labeled containers for these substances.
- Provisions for spill prevention, response, and disposal of wastes generated from any potential clean-up activities from Contractor chemicals / hazardous materials must be discussed with the JPL contact before starting work with these substances.

11.0 PERSONAL PROTECTIVE EQUIPMENT (PPE)

- **General:** "PPE Hazard Assessment" means the process of identifying, selecting, and documenting appropriate personal protective equipment (PPE) for workplace hazards.
- Contractor shall provide, at its own expense, and enforce the use of all appropriate job specific PPE's and any posted Company requirements.
- Contractor shall ensure that all workers are trained in the proper use of applicable PPE's prior to performing Work.
- Approved hard hats, hard-toed boots or shoes, and safety glasses shall be worn on all Company worksites.
- Loose or floppy clothing, neck chains, loose jewelry, or loose long hair is prohibited.
- Rings shall be removed when working in areas where they could catch on moving objects or sharp protrusions or come into contact with electrical circuits.
- Clothing, including gloves, shall not be cleaned by blowing with compressed air, blowing with compressed gases, or washing in a flammable liquid

TO HAVE UNIFORMITY IN THE USAGE OF PPE'S, APPROVED MODELS AND MAKE OF PPE'S AS MENTIONED BELOW ARE ACCEPTABLE AT JPL WORKPLACE.

LIST OF PPES & SAFETY EQUIPMENT'S (APPROVED MAKE/MODEL)			
S.No.	Item	Make/Model	Photograph
1	Industrial Safety Helmet	Joseph leslico / Karam/ 3M/Savior make with ratchet. IS 2928 & EN 397 approved Qty: White for Engineer / Supervisor, Yellow for Workman, Red for Electrician, & Green for Safety.	
2	Safety Shoes	LIBERTY make "WARRIOR" brand / Allen Cooper / Karam / Udyogi/ Bata low-cut 2mm thick leather safety shoes with IMPORTED Esjot alloy toecaps & Double density directly injected PU black & Grey sole/ Safety shoe can allow ISI mark. As IS:15298	
3	Ear Plug	Honeywell / Karam/ 3M™ E-A-R™ UltraFit™ cloth/cotton Corded Earplugs, in poly bags/Carrying Case 2. Reusable, Triple flange design. NRR 25 dB	

		Karam EP 02 / 3M 32,dB	
4	Ear muff	Leslico / Karam /3M Hard hat mounted 1450 ear muff- NRR 23 dB	
		Helmet attachable ear muff EP 23, NRR 27 dB, EN 352- 1; 2002	
5	safety spectacles (shaded/clear)	Karam / 3M™ Virtua™ / UVEX Protective Eyewear, 11326-00000-100 Clear Temples Clear Hard Coat Lens 100 ea/case 3M™ Virtua™ Protective Eyewear, 11327-00000-20 Gray Hard Coat Lens, Gray Temple 20 ea/case Karam/3M™ Virtua™ / UVEX Protective Eyewear, 11329-00000-20 Clear Anti-Fog Lens, Clear Temple 20 ea/case Kara / 3M™ Virtua™ / UVEX Protective Eyewear AP, 11815-00000-20 Gray Hard Coat Lenses 20 ea/cs' & 3M™ Virtua™ Protective Eyewear AP, 11819-00000-20, Clear Hard Coat Lenses 20 ea/cs'	 
6	Goggles (for chemical handling)	Karam make / UVEX/ 3M etc. ES 009 clear, Confirms to EN 166:2001 Karam / UVEX / 3M 1621 safety goggles for splashes (can be worn over prescription lens). ANSI Z87.1-2003	
7	Respirators (dust mask)	3M / Leslico / Venus 8210 dust respirator, N 95 , NIOSH approved, ISI Approved.	
8	Welding Respirators	3M / Leslico / Venus welding respirator 8512, NIOSH approved	
9	Half face mask	3M / Venus half face piece reusable respirator 6200 series (medium size)(to be used with cartridges),NIOSH approved	

		Honeywell / 'Saviour' Electrex Model No.– ELECTREX-33, ERDA Approved-Tested as per IS: 4770, 1991 (marketed by Suresafety), Provides protection from 33 KV Test voltage Electrical operations	
		Honeywell / 'Saviour' Electrex Model No.– ELECTREX-11, ERDA Approved-Tested as per IS: 4770, 1991 (marketed by Suresafety), Provides protection from 11 KV Test voltage Electrical operations	
17	Full Body Safety harness	Karam / Unicare / Udyogi brand full body harness model: Rhino PN 23 with PN 351 Double Lanyard Scaffolding hook and energy absorber. Conforms to EN 361: 2002, CE approved.	
18	Retractable wire rope fall arrester	Karam / Unicare/ Udyogi brand, PCGS 15, PCGS 20, PCGS 25. EN 360	
19	Fall arrester with energy absorber	Karam / Unicare / Udyogi PN 2000 B. EN 353	
20	Anchorage webbing slings	Karam / Unicare / Udyogi brand Concrete anchor strap PN 805 and PN 806 (lengths 1.0m, 1.5 m, 2.0 m). EN 795	
21	Anchorage SS wire rope	Anchorage SS wire rope Karam PN 814. EN 795 or Unicare / Udyogi	
22	Safety net	Karam / Garware/ Udyogi / Safetech make Safety net made from Polypropylene ropes. Mesh size:- 25 mm & 100 mm, Size:- 10 m X 5 m, mesh rope:- 2mm/4mm double cord, with overlay net	
23	Temporary horizontal lifeline	Karam / Unicare / Udyogi PN 3000, EN 795 Class B, made up of 30 mm polyester webbing and ratchet tensioner. Both ends fitted with auto locking steel karabiners.	
24	Vertical lifeline	Karam / Unicare / Udyogi, 3 strand polyamide twisted rope of dia 14 mm, one end loop and other end stop knot. 10 m to 200 m (PN 910 to PN 9200)	

25	Work positioning lanyard	Karam / Unicare / Udyogi make work positioning lanyard with ring type adjuster PN 241 . Made of 14 mm dia polyamide 3 strand twisted rope. Steel karabiner PN 112 at both ends. Manual ring type adjuster. EN 358	
26	Fire blankets	Udyogi/ Karam etc. make Fire blanket compact, wall mounted, easy to use made of asbestos free Material in size : 1.2mtr X 1.8mtr	
27	Gas welder's glass	Karam ES 003 / Honeywell/ Unicare fitted with IR 5 lens. EN 166	 ES 003 (IR 5)
28	Arc welder's glass	Karam / Honeywell / Unicare ES 004 fitted with IR 11 lens. EN 166	 ES 004
29	Helmet attachable welding shield	Karam / Honeywell make ES 71. It fits standard helmet with 30 mm slot. Confirms to EN 175 and ANSI Z 87.1. (To be fitted with IR 5 / IR 11 lens, sandwiched between two polycarbonate lens, confirming to EN 166 and ANSI Z 87.1)	 Ref. ES 71
30	Helmet attachable grinding shield	Karam ES 51 (02)/ Honeywell or Joseph Leslico/ equivalent. CE certified	 Ref. ES 51
31	Eye Wash Bottle	Unicare make UEWB 12 / Udyogi EW/ equivalent – 500 ML	
32	Barricading tape	Made up of virgin quality pvc material tubing pattern. Roll red & white colour with "DANGER/STOP & CAUTION/WORK IN PROGRESS" letters in Hindi &English Size : 3" Make : SAFE-T-PLUS / PRIMA or equivalent	
33	Life buoy	Unicare or equivalent The Life Buoy is manufactured in durable high visibility orange synthetic material and requires no maintenance. Moisture proof and non-inflammable. The H-Glow reflective material fixed in the indentations identifies victims in distress. Approved by the Mercantile Marine Department to SOLAS spec.	

34	Life jacket	Unicare or equivalent •MMD approved to Solas Specifications •With Solas Reflective Tape, Signaling Whistle and Light Quick turning time and high free board in water •Buoyancy : 17.5 Kg •Light that is automatically activated when in contact with sea water/ petroleum products. •Size : 80 X 37 X 10cms. Approved by the Mercantile Marine Department to SOLAS spec.	
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12.0 ELECTRICAL SAFETY

- Qualified Person – A designated worker who has the skills and knowledge related to operation of the electrical equipment and installations to be employed who should have received training to recognize and avoid the electrical hazards involved. Usually, this is a licensed electrician or someone with certain equivalent experience and training. A person can be “qualified” to work with certain equipment and methods and still be “unqualified” for other work.
- Contractor shall perform all electrical Work in accordance with the current editions of applicable central, state and local laws, rules, regulations, and standards.
- Installation of electrical systems or modifications to electrical systems shall be done under the supervision or direction of a licensed electrician.
- Contractor shall ensure that workers near overhead power lines know the voltage of the line and the safe approach distance.
- Contractor shall be aware of, and take precautions to prevent, the build-up of static electricity in areas with a potential Hazardous Atmosphere.
- Only qualified Person shall discharge all stored electrical energy and shall verify that equipment is de- energized and proper Lockout/ Tagout (LO/TO) procedures have been implemented prior to beginning electrical Work.
- All power lines shall be considered energized unless proper measures have been taken to de-energize.

13.0 LOCKOUT/TAGOUT (LOTO)

- Contractor shall ensure compliance with all requirements of company LOTO procedure.
- Contractor shall ensure that its workers are adequately trained in LOTO and applicable energy control procedures.
- LOTO energy control procedures shall be followed prior to work on any equipment or process where stored energy or the unexpected energizing of equipment could cause injury to a worker. Potential energy sources include electrical, mechanical, pneumatic, hydraulic, thermal, chemical, natural gas, and all forms of potential and stored energy.
- Repairs, maintenance, or alterations shall not be made on equipment in operation. All equipment shall be shut down and a LOTO device placed in a manner that the equipment cannot be accidentally started.

- Contractor shall ensure that a briefing is conducted with all workers affected by a LOTO operation before each shift, and more frequently if warranted by personnel changes or changes in the scope of Work. The briefing should include the following items:
 - i. The specific equipment or process involved, along with any related equipment;
 - ii. The estimated length of time required to complete the task;
 - iii. The hazards involved in performing repairs or maintenance, including the potential hazards to workers if the equipment or process is prematurely energized; and
 - iv. A review of the site-specific energy control procedure.
- To ensure the machine or equipment has been properly locked out of service prior to starting any Work, a Qualified Person shall attempt to turn on the power source to verify that the machine or equipment does not become energized.

14.0 HOT WORK

- "Hot work" means riveting, welding, flame cutting or other fire or spark-producing operation. No hot work is to be performed without first obtaining an JPL Hot Work permit, except in designated locations including Workshop and facilities Maintenance areas.
- Your JPL contact will help you identify the need for and to complete required permit requirements.

15.0 COMPRESSED GAS SAFETY

- Compressed gas cylinders must be secured in an upright position and kept away from sources of heat or flame at all times.
- All compressed gas cylinders must be legibly marked with either their chemical or trade name.
- All compressed gas cylinders not in use must have their top caps securely tightened.
- Where different gases are stored, they must be grouped by types. Groupings shall separate the flammable gases from the oxidizing gases.
- All oxygen cylinders must be stored not less than 20 feet from fuel gas cylinders or other flammable gasses.
- Compressed gas cylinders may not be dropped, dragged, rolled on their side or struck violently.
- A compressed gas cylinder may only be lifted by cradles or enclosed platforms when using a crane or hoisting device.
- A frozen or ice-clogged valve shall be thawed either by warm air or use of warm water and dried before using. Boiling water or a flame shall not be used. Force shall not be applied to a valve or cap to loosen a cylinder frozen in place.

- A cylinder shall not be placed where it will become a part of the electrical circuit by accidental grounding or where it may be burned by electric welding arc. A cylinder shall not be placed so that hot slag or flame will reach it or it shall be protected by a fire resistant shield.
- Flammable gas cylinders including LPG cylinders may not be stored within 25 feet of an emergency exit.
- Flashback arrestors/preventers are required on all cylinders and torches.

16.0 CONFINED SPACES

- Confined space entry requires an JPL Confined Space Entry Permit. The Contractor is responsible for compliance with the standards for any entry by a Contractor Employee.
- Confined space entry training is also a prerequisite for any entrants or attendants as part of this program.
- Entry equipment must be provided by the Contractor and calibration information must be available upon request.
- Emergency rescue procedures, equipment and personnel are the responsibility of the Contractor.
- Entry / Exit attendance format to be filled.

17.0 LADDERS

PORTABLE LADDERS:

- Contractor shall ensure that all workers have been trained in the proper use, placement, care and maximum load carrying capacities of the ladders used.
- Contractor shall inspect all ladders before use. Any damaged or unsafe ladders shall be tagged and taken out of service.
- Portable ladders shall be set at the correct angle. The distance from the foot of the ladder to the structure should be equal to 1/4 the length of the ladder.
- Workers shall keep both hands free for climbing, descending, and performing Work on a ladder. Carrying hand tools or equipment while climbing on a ladder is prohibited unless secured in a pocket or on a belt. Articles that are too large to be carried in a pocket or on a belt shall be lifted and lowered by a hand line.
- Workers shall not rush, but shall take one step at a time and face the ladder while climbing and descending.
- Only one person at a time shall be on the ladder.
- Portable ladders shall have anti-slip safety feet and be secured at the top before work begins in order to prevent the ladder from shifting. A second person shall hold the ladder until the climber can secure it at the top.
- Ladders shall extend at least 3 feet above the point of support when gaining access to a roof or other area.
- Contractor shall use fall protection on ladders when additional significant hazards such as impalement, rotating machinery, or electrical shock are present.

- Ladders shall be maintained free of oil, grease, and other slipping hazards.
- Workers shall maintain their body's center of gravity between the side rails at all times while working from a ladder. In addition, workers shall avoid work from a ladder that involves significant pushing, pulling, or any action that may dislodge the person from the ladder.
- The top two steps of a step ladder shall not be used as steps. This requirement does not apply to step ladders with three steps or less or to step ladders with a guard rail-equipped work platform at the top.

18.0 SCAFFOLDING:

- Scaffolds or elevated platforms shall be constructed, maintained, and used in accordance with the applicable regulations and company standards.
- Contractor shall ensure that a Competent Person is assigned to supervise scaffold erection, dismantling, alteration, and movement.
- Contractor shall ensure that all scaffold materials and planking are thoroughly inspected for defects prior to use.
- Where there is a hazard to workers working below an elevated scaffold, toe boards shall be in place.
- Climbing or working from the handrail, mid-rail, or brace members of the scaffolding is prohibited.

19.0 FALL PROTECTION

- Definitions:
 - a) Fall Protection Competent Person – A person who is knowledgeable of:
 - b) The fall hazards at the worksite;
 - c) Correct procedures for assembling, maintaining, disassembling, and inspecting fall arrest equipment; and
 - d) The operation of guardrail systems, Personal Fall Arrest Systems (defined below), warning line systems, safety monitoring systems, and other protection to be used.
 - e) Personal Fall Arrest System – A system used to arrest a person in a fall from a working level at height. It consists of an anchorage, connectors, and body harness. It may include a lanyard, deceleration device, lifeline, or combination of these.
- A Personal Fall Arrest System shall be worn:
 - a) On work surfaces where potential drop is greater than 6 feet (1.8 meters).
 - b) When working on unfinished structures greater than 6 feet (1.8 meters) in height where the work surface is without guardrails, toe boards, or gated access ladders;

c) When working on areas within 6 feet (1.8 meters) of the edge of a work surface greater than 6 feet in height or within 6 feet of any unguarded opening, skylight, service duct, stairwell, or elevator shaft on a roof or unfinished level of a structure;

d) When working along unguarded locations at the edge of a well, pit, shaft, excavation, trench, or similar location 6 feet or more in depth when the excavation is not readily seen because of plant growth or other visual barrier;

e) Whenever deemed necessary by a safety officer.

- Contractor shall inspect all components of a Personal Fall Protection System before each use and replace if necessary. Fall protection equipment that has been subjected to impact loading shall be immediately removed from service and shall be inspected by the manufacturer before reuse or destroyed and replaced.
- Contractor shall ensure that components of a Personal Protection System are free from defect such as cuts, tears, abrasions, mold, undue stretching, missing or degraded stitching, alterations, or additions that might affect its efficiency. Contractor shall also inspect for damage due to chemical exposure, deterioration, distorted hooks, or faulty hook springs, loose or damaged mountings, non-functioning parts, wearing, or internal deterioration in the ropes or webbing.
- Contractor shall follow the manufacturer's recommended procedures for fitting, adjusting, using, inspecting, testing, and caring for fall protection equipment.
- A Personal Fall Arrest System shall not be used as a primary suspension device for positioning, or as a retrieval system, or for transporting materials.
- Contractor shall evaluate the compatibility of all fall arrest systems and anchorage points prior to use.
- Contractor shall calculate tie-off distances accurately to limit a fall to a maximum of 6 feet, considering lanyard elongation, Work position, proximity to fall area, and the location of fall hazards. The anchorage and tie-off points should be located to avoid obstructions in the potential

20.0 CHEMICAL SAFETY

- JPL uses a variety of chemicals that Contractors may encounter. JPL maintains MSDS's for these chemicals. Contractors may request a copy of any/all MSDS's for chemicals to which they are (or may be) exposed by contacting their JPL contact.
- Use of JPL chemicals by a Contractor for any purpose must be authorized in advance by the JPL contact.
- A current MSDS must be readily available and maintained by the Contractor for all chemicals brought onto JPL premises.

21.0 HAND TOOLS

- All hand tools, including portable electrical tools, and other equipment brought onto JPL premises must be in safe condition. JPL reserves the right to prohibit the use of defective tools, ladders, etc. and dictate the removal of defective equipment.

22.0 CRANES AND HOISTS

- JPL owned Overhead Cranes and Hoists are not to be operated by Contractors unless they receive advance authorization from the JPL contact.
- Upon authorization, only Contractor Employees for whom the Contractor has provided training sufficient to meet the standards for cranes and hoists (including applicable licensing) will be allowed to operate this equipment.
- Contractor owned Cranes and Hoists must be operated under the requirements of Safety Standards for Overhead and Gantry Cranes and Monorail Systems. This includes training/licensing requirements, equipment inspection, and safe equipment operation requirements.
- Any crane, hoist or rigging system that is not safe to operate must be tagged out of service and not operated until repaired/serviced.

23.0 HOUSEKEEPING

- Job site housekeeping is the contractor's responsibility and the job site must be as clean and orderly as possible while work is being performed. Good housekeeping practices are of utmost importance in preventing injuries.
- DO NOT LITTER. Contractors are responsible to keep the work area(s) in a neat and orderly condition at all times. All material must be cleaned up as the job progresses.
- All roadways, passageways, and operating areas must be kept clear at all times

24.0 WASTE DISPOSAL

- Proper identification of waste generated during work on-site is critical. All wastes must be segregated and managed according to applicable regulatory requirements.
- The Contractor is responsible for the removal of any waste generated.
- It is the responsibility of the Contractor to ensure proper waste management practices while performing services for JPL. Prior to any work, the Contractor will assess what wastes will be generated and communicate to the JPL contact any hazardous, non-hazardous, universal or construction wastes that will be generated and the methods that will be used to collect, manage, and dispose of these wastes.
- Discharge of any material onto the ground is strictly prohibited by State/National Laws and JPL HSE policy. If any spill/discharge occurs, report it immediately to your JPL Contact (i.e. leak from truck of oil, gas, or product being transported).
- Any questions regarding proper classification and/or disposal of wastes generated must to be brought to the attention of your JPL contact.

25.0 ROOF WORK

- The portion of a roof where Contractors are working must be kept free of slippery conditions.
- All roof work performed on JPL premises must be conducted in accordance with Safety Standard for Fall Protection.

26.0 RIGGING

- All rigging must be done in accordance with applicable regulations.

27.0 OVERHEAD WORK

- Contractor shall ensure that workers are trained to recognize the hazards of working around overhead utility lines and how to minimize the chance of contact.
- Contractor shall take precautions to ensure the safety of workers and ensure the integrity of the existing overhead utility lines.
- Contractor shall conduct a hazard assessment to identify and mitigate hazards prior to working around overhead utility lines. The hazard assessment shall include the following:
 - Identifying all overhead utility lines (on or off the worksite) that may be impacted by the Work;
 - Verifying that appropriate signage and visual warning devices are installed to alert workers to the hazards;
 - Clearly marking or otherwise restraining all lifting or boom-type equipment to show the maximum height or extension possible as measured from ground level or to limit the maximum limit of extension, respectively; and
 - Using adequately trained and dedicated spotters at locations where equipment and vehicles pass or work under or around utility lines.
- Physical barriers: Non-conductive, highly visible devices (e.g., goal posts, barricade tape) set outside the limits of approach (limits shall vary by jurisdiction, land restriction, and voltages) on both the coming and going away sides.
- Site-specific controls prescribed and authorized for use by the utility owner (e.g., shielding, de-energizing, bonding, insulating).
- Keep all equipment attachments in the lowest possible position when traveling under overhead utilities.
- Use dry tag lines made of a nonconductive type material when working near energized lines.
- All ladders used around power lines shall be made of non-conductive materials;
- Use only non-conductive chokers, slings, and lifting devices during material handling activities;
- Keep materials bonded at all times when transporting conductive loads, (e.g., pipe, air compressor, pumps) in the proximity of high voltage lines.

- All overhead work must be conspicuously barricaded or otherwise marked to prevent anyone from walking or driving under the work area.
- Overhead work creating sparks requires a Hot Work Permit.
- Overhead work creating falling debris requires additional protection for personnel and equipment that may be affected by the falling debris
- All scaffolds must have full flooring within the frame, with cleats, toe boards, and railings and meet BIS requirements.

28.0 ELEVATED WORK

- General: When working overhead, Contractor shall protect people below. Contractor shall ensure that tools, materials, and equipment subject to falling from height are adequately secured before Work is performed. Tools and materials shall be handed up or down, but never thrown. When it is necessary to hoist tools with a rope, exercise care to ensure the tools are securely attached to the line or loaded into a container and there is no danger of them being dropped.

29.0 WORK ZONE TRAFFIC CONTROL

- If a Work activity is conducted on or near a road, Contractor shall comply with all applicable regulations.
- Contractor shall provide all signs, barriers, flaggers, and other notification necessary to protect its workers and the public from damage, injury, or loss. Barricades at public areas (e.g., road crossings) shall have flashing lights/ reflective during hours of darkness.
- All work conducted in on or near a road at night requires the use of high visibility traffic vests.

30.0 PENALTY SYSTEM

- On non-compliance of PPEs and other safety instructions, following penalties will be imposed on the contracting agency as per below mentioned violations. The amount towards the violation shall be deducted from monthly bill of the contracting agency.

S.No.	Violation	Penalties
1	Non-use of any of PPE like Safety Helmet with chin strap, Safety shoes by individuals	Lapses & non-compliance in safety related issues in repetitive manner will be viewed very seriously by JPL Management and further a monetary amount as deemed fit by JPL's Management will be debited from your Bill without any consideration.
2	Non-use of Full body Harness at height	
3	Repetition of violation by employees of same contractor within a week	
4.	Over speeding or wrong side driving	
5.	Non reporting of accident	
6.	Working without work permit	

7.	Non-use of proper tools & tackles i.e. Cutting torch without Flash back arrestor at both ends, grinder without Guard, Lifting tools and tackles without certification etc.	
8	Allow to work on visitor pass	
9	Overloading during material handling	
10	Smoking at workplace	
11	Non-use of PPE's as per the job requirement.	

IN CASE AGENCY OR INDIVIDUAL IS A FREQUENT DEFAULTER, A RED MARK SHALL BE RECORDED IN HIS PERFORMANCE RECORD & THE AGENCY SHALL BE BARRED AND BLACK LISTED FOR REPETITIVE NON COMPLIANCES ON FRONT OF SAFETY.

31.0 ACCIDENT REPORTING AND INVESTIGATION

- The Contractor must immediately report any injury, illness, or near miss that occurs at JPL premises to the JPL contact.
- The information related to the incident must be provided to JPL contact as per the JPL Incident reporting and investigation Procedure.

32.0 GENERAL RULES TO FOLLOW

- Always store materials in a safe manner. Tie down or support piles if necessary to prevent falling, rolling, or shifting.
- Dust scraps, oil or grease should not be allowed to accumulate. Good housekeeping is a part of the job.
- Trash piles must be removed as soon as possible. Trash is a safety and fire hazard.
- Remove or bend over the nails in lumber that has been used or removed from a structure.
- Immediately remove all loose materials from stairs, walkways, ramps, platforms, etc.
- Do not block aisles, traffic lanes, fire exits, gangways, or stairs.
- Avoid shortcuts – use ramps, stairs, walkways, ladders, etc.
- Standard guardrails must be erected around all floor openings and excavations must be barricaded. Contact your supervisor for the correct specifications.
- Do not remove, deface or destroy any warning, danger sign, or barricade, or interfere with any form of protective device or practice provided for your use or that is being used by other workers.
- Get help with heavy or bulky materials to avoid injury to yourself or damage to material.
- Keep all tools away from the edges of scaffolding, platforms, shaft openings, etc.
- Do not use tools with split, broken, or loose handles, or burred or mushroomed heads. Keep cutting tools sharp and carry all tools in a container.
- Know the correct use of hand and power tools. Use the right tool for the job.
- Know the location and use of fire extinguishing equipment and the procedure for sounding a fire alarm.
- Proper guards or shields must be installed on all power tools before use. Do not use any tools without the guards in their proper working condition. No “homemade” handles or extensions (cheaters) will be used!
- All electrical power tools (unless double insulated), extension cords, and equipment must be properly

grounded.

- All electrical power tools and extension cords must be properly insulated. Damaged cords must be replaced.
- Do not operate any power tool or equipment unless you are trained in its operation and authorized by your firm to do so.
- All electrical power equipment and tools must be grounded or double insulated.
- Use tools only for their designed purpose.